

PERMUTATION & COMBINATION-1

PERMUTATION AND COMBINATION-1

1. 25 buses are running between two places Punjab and Haryana. In how many ways can Sunil go from Punjab to Haryana and return by a different bus?

- A. 600
- B. 625
- C. 576
- D. None of these

PERMUTATION AND COMBINATION-1

2. There are 10 women and 15 men in coffee shop. In how many ways can a person can be selected?

- A. 25
- B. 30
- C. 15
- D. 50



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3. In how many different ways can the letters of the word 'RUMOUR' be arranged?

- (A) 180
- (B) 360
- (C) 540
- (D) 620

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4. In how many ways can you rearrange the word 'JUMBLE' such that the rearranged word starts with a vowel?

- A) 120
- B) 240
- C) 360
- D) 60

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5. How many different words can be formed with the letters of the word 'FAMILY' when vowels occupy even places?

- A) 18
- B) 36
- C) 72
- D) 144

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6. In how many different ways can the letters of the word 'BANKING' be arranged so that the vowels always come together?

- (A) 220
- (B) 260
- (C) 450
- (D) 720



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7. How many words can be formed from the letters of the word 'SIGNATURE' so that the vowels always come together?

- (A) 750
- (B) 1450
- (C) 2880
- (D) 17280

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8. In how many different ways can the letters of the word 'AUCTION' be arranged in such a way that the vowels do not come together?

- (A) 4464
- (B) 5547
- (C) 1650
- (D) 5765



PERMUTATION AND COMBINATION-1

9. Out of 7 consonants and 4 vowels, how many words of 3 consonants and 2 vowels can be formed?

- (A) 320
- (B) 450
- (C) 25200
- (D) 15920

PERMUTATION AND COMBINATION-2

10. A letter lock consists of three rings each marked with six different letters. The number of distinct unsuccessful attempts to open the lock is at the most ?

1. 250
2. 100
3. 150
4. 215

PERMUTATION AND COMBINATION-2

11. The number of ways that 8 beads of different colours be strung as a necklace is

1. 2520
2. 2880
3. 4320
4. 5050

PERMUTATION AND COMBINATION-2

12. How many necklace of 12 beads each can be made from 18 beads of different colours?

1. $18! \times 19!$
2. $18! \times 30$
3. $18!$
4. $18!(6 \times 24)$



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13. How many two digit numbers can be generated using the digits 1,2,3,4 without repeating any digit?

- A. 10
- B. 12
- C. 4
- D. 16

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14. How many four digits numbers can be formed with the digits 0, 1, 2, 3, 4, 5, 6 and 7; digits being used more than once?

- A) 4000
- B) 4220
- C) 3584
- D) 2100



PERMUTATION AND COMBINATION-1

15. How many numbers of four digits greater than 2,400 can be formed with digits 0, 1, 2, 3, 4, 5 & 6; no digit being repeated in any number?

- A) 140
- B) 480
- C) 540
- D) 1120



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16. How many three digit numbers can be generated using the digits 1,2,3,4 and 5 divisible by 2 without repeating any digit?

- A. 20
- B. 26
- C. 40
- D. 24

PERMUTATION AND COMBINATION-1

17. How many three digit numbers can be generated using the digits 1,2,3,4 and 5 divisible by 4 without repeating any digit?

- A. 30
- B. 12
- C. 20
- D. 24

PERMUTATION AND COMBINATION-1

18. How many three digit numbers can be generated using the digits 2,3,5,6 and 7 divisible by 5 without repeating any digit?

- A. 30
- B. 12
- C. 20
- D. 24

PERMUTATION AND COMBINATION-2

19. A box contains three white balls, four black balls and three red balls. The number of ways in which three balls can be drawn from the box so that at least one of the balls is black is

1. 50
2. 100
3. 150
4. 200

PERMUTATION AND COMBINATION-2

20. In how many ways can 5 boys and 4 girls be seated in a row, so that they alternate?

- A) $5!$
- B) $5! \times 2!$
- C) $4! \times 5!$
- D) None of these

PERMUTATION AND COMBINATION-2

21. In how many ways can 5 boys and 5 girls can be seated in a row so that boys and girls are placed alternately?

- A) $5!$
- B) $5! \times 2!$
- C) $2 \times 5! \times 5!$
- D) None of these

PERMUTATION AND COMBINATION-2

22. There are 5 yellow, 4 green and 3 black balls in a bag. All the 12 balls are drawn one by one and arranged in a row. Find out the number of different arrangements possible.

- A. 25230
- B. 21200
- C. 23420
- D. 27720



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23. In how many ways can a team of 5 persons be formed out of a total of 10 persons such that two particular persons should not be included in any team?

- A. 112
- B. 128
- C. 56
- D. 28



PERMUTATION AND COMBINATION-1

24. In how many ways can a group of 5 men and 2 women be made out of a total of 7 men and 3 women?

- (A) 63
- (B) 98
- (C) 65
- (D) 57

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25. Find the sum of all the 4 digit numbers that can be formed with the digits 3, 4, 5 and 6

- A) 119988
- B) 11988
- C) 191988
- D) None of these

PERMUTATION AND COMBINATION-1

26. Find the sum of all the 4 digit numbers that can be formed with the digits 3, 4, 4 and 2.

- A) 43339
- B) 43999
- C) 43329
- D) None of these



PERMUTATION AND COMBINATION-1

27. In a group of 6 boys and 4 girls and 4 children are to be selected. In how many different ways can they be selected such that at least one boy should be there?

- A. 209
- B. 150
- C. 501
- D. 250

PERMUTATION AND COMBINATION-1

28. In how many ways a committee, consisting of 5 men and 6 women can be formed from 8 men and 10 women?

- (A) 61300
- (B) 9810
- (C) 23500
- (D) 11760



PERMUTATION AND COMBINATION-1

29. If there are 9 horizontal lines and 9 vertical lines in a chess board, how many rectangles can be formed in the chess board?

- A. 1024
- B. 64
- C. 1296
- D. 920

PERMUTATION AND COMBINATION-1

30. In an examination, a candidate is required to pass all five different subjects. The number of ways he can fail is:

- A) 32
- B) 31
- C) 30
- D) 29

Any Doubts???